

UFPR-MoTri: Fine-Grained Type Recognition of Two- and Three-Wheeled Vehicles in Surveillance

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Introduction

Vehicle Type Recognition is crucial for Intelligent Transportation, yet two- and three-wheeled vehicles remain underexplored in surveillance. Furthermore, legal type taxonomies fail to capture visual characteristics (Fig. 1), hindering accurate recognition.

Proposal: We introduce UFPR-MoTri, a public dataset of two- and three-wheeled vehicles, and benchmark six deep learning models using a proposed visually grounded taxonomies.

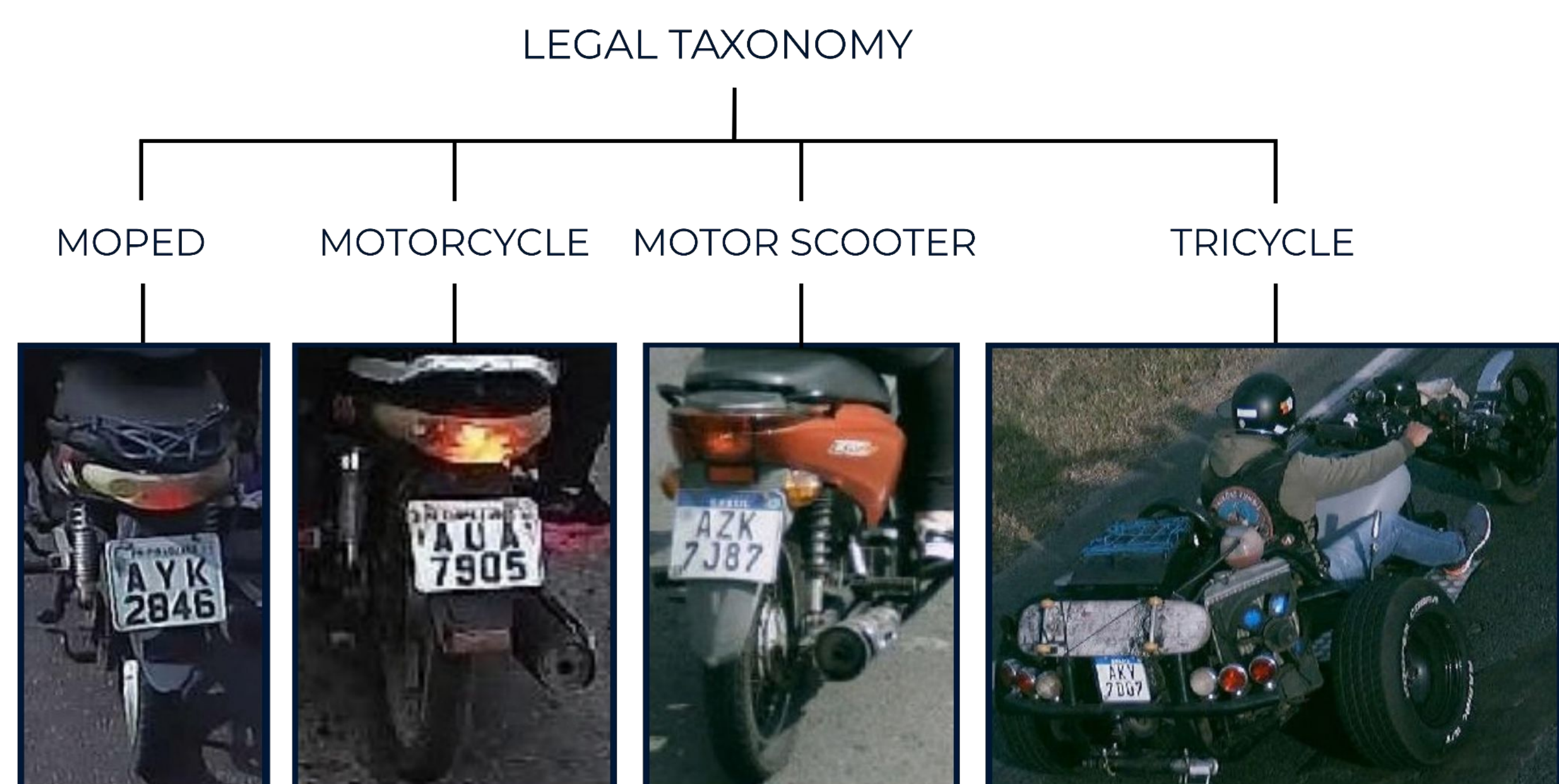


Fig. 1. Brazilian Legal Type Taxonomy.

UFPR-MoTri Dataset

Dataset main characteristics:

- 94,083 rear-view images from 41,838 unique vehicles;
- Images provided by the Military Police of Paraná;
- High diversity of scenarios, illumination, viewpoints, and vehicle modification (Fig 2.).



Fig. 2. Samples from the UFPR-MoTri dataset.



The dataset is available under request. Scan for details and access instructions on GitHub.

Taxonomy

Brazilian Legal Taxonomy (4 classes): Baseline relying on administrative criteria.

Fine-Grained Taxonomy (12 classes): Proposed scheme based on market segments.

Superclass Taxonomy (6 classes): Proposed scheme grouping visually similar vehicles, aiming to mitigating ambiguity (Fig. 3).

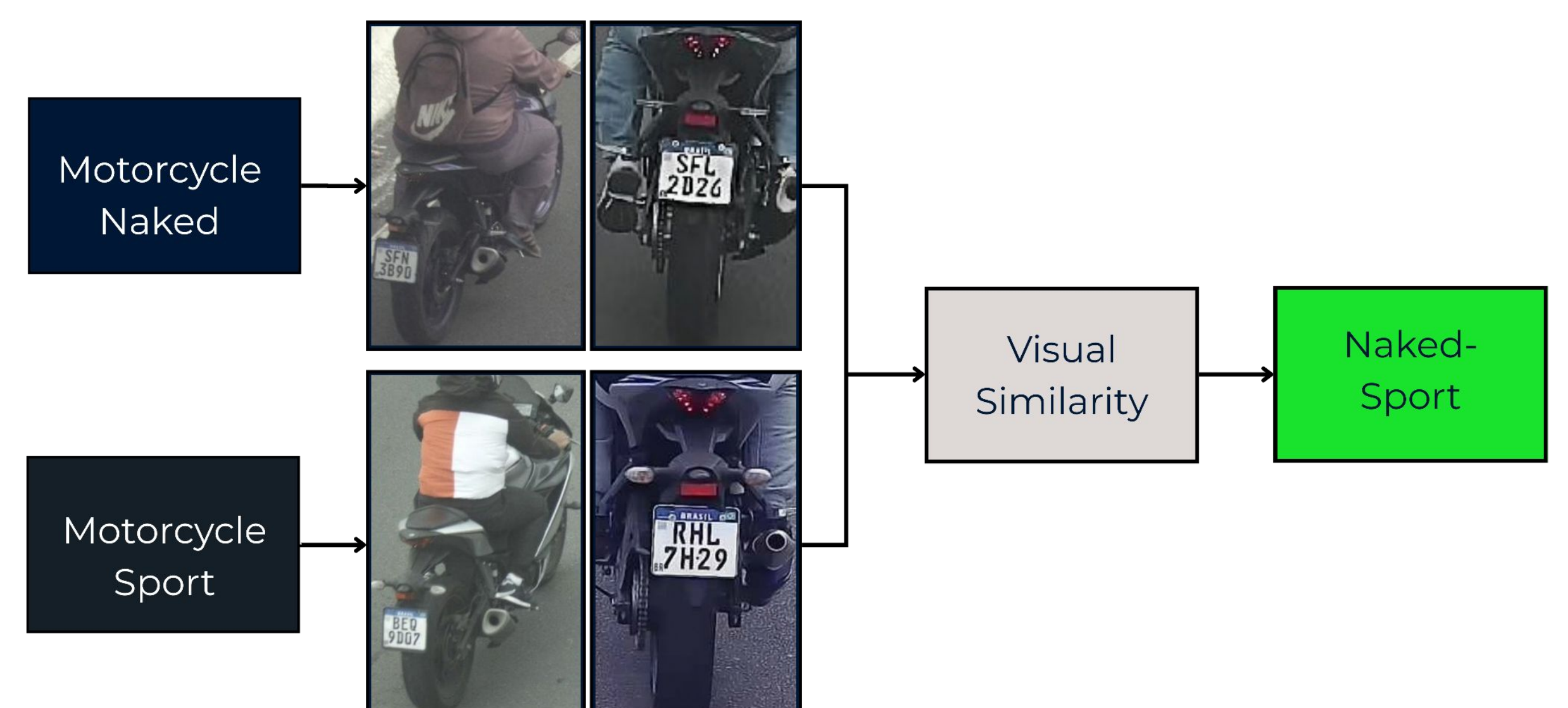


Fig. 3. Example of class grouping based on visual similarity.

Experimental Setup and Results

We evaluated and compared six deep learning architecture under the three previously defined type taxonomies.

Model	Legal Type		Fine-grained Taxonomy		Superclass Taxonomy	
	Mi-Acc	F1	Mi-Acc	F1	Mi-Acc	F1
DINOv2	98.4	77.5	96.6	75.1	98.0	92.5
ConvNext-Tiny	98.9	72.8	96.6	65.7	97.6	91.5
EfficientNet-V2-Small	98.9	84.6	95.7	64.3	97.1	89.9
CoAtNet	98.7	76.9	96.2	64.7	97.2	90.5
ResNet-50	97.4	68.8	94.1	62.0	95.2	84.9
SwinTransformer-Tiny	98.6	77.9	95.1	63.4	96.7	88.8

Table 1. Classification results across the three evaluated schemes.

Key Findings:

- Rear-view captures obscure the details necessary for a finer-grained taxonomy.
- The superclass taxonomy reduces ambiguity, achieving results of 98.0% Micro-Accuracy and 92.5% Macro-F1.

Conclusion

The proposed visual superclass taxonomy provides a more reliable recognition framework than standard administrative labels. Future work will extend this evaluation to recognize the makes and models of two- and three-wheeled vehicles.

Acknowledgments

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